Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- Convex Optimization
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- **Quadratic Programming**
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Quadratic Programming

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Quadratic Programming (QP) is an extension of linear programming where the objective function is quadratic and the constraints remain linear. Quadratic programming problems are a subset of convex optimization problems, and they appear in various applications, including finance, machine learning, and control systems.

Formulation of a Quadratic Programming Problem

A standard quadratic programming problem can be formulated as:

$$\text{Minimize} \quad \frac{1}{2}x^T Qx + c^T x$$

$$\text{Subject to} \quad Ax \leq b$$

\begin{align*}x &\in \mathbb{R}^d \quad \text{is the vector of decision variables,} \\Q &\in \mathbb{R}^{d \times d} \quad \text{is a symmetric matrix defining the quadratic term,} \\c &\in \mathbb{R}^d \quad \text{is the vector defining the linear term,} \\A &\in \mathbb{R}^{m \times d} \quad \text{is the matrix representing the linear constraint coefficients,} \\b &\in \mathbb{R}^m \quad \text{is the vector representing the right-hand side of the constraints.}\end{align*}

The goal in quadratic programming is to find the vector x that minimizes the quadratic objective function while satisfying the linear constraints.

Objective Function

The objective function in a quadratic programming problem has two components:

- The quadratic term $\frac{1}{2}x^T Qx$: This term defines how the objective function curves, with Q being a symmetric matrix. If Q is positive semidefinite (all its eigenvalues are non-negative), the function is convex, ensuring that any local minimum is also a global minimum.
- The linear term $c^T x$: This term represents a linear contribution to the objective function, similar to the objective in linear programming.

Constraints

The constraints in quadratic programming are typically linear, of the form $Ax \leq b$, similar to those in linear programming. These constraints define a convex feasible region, which is the set of all points x that satisfy the constraints.

Lagrangian and Dual Problem

As with linear programming, quadratic programming problems can also be approached using Lagrange multipliers to form a dual problem. The Lagrangian for the quadratic programming problem is:

$$L(x, \lambda) = \frac{1}{2}x^T Qx + c^T x + \lambda^T (Ax - b)$$

Here, $\lambda \in R^m$ are the non-negative Lagrange multipliers associated with the constraints $Ax \leq b$.

The dual problem is derived by first taking the derivative of the Lagrangian with respect to x and setting it to zero, which yields:

$$Qx + c + A^T \lambda = 0$$

This condition allows us to express x in terms of λ:

$$x = -Q^{-1}(c + A^T \lambda)$$

Substituting this into the Lagrangian gives the dual objective function $D(\lambda)$, and the dual problem becomes:

$$\text{Maximize } D(\lambda) = -\frac{1}{2}(c + A^T \lambda)^T Q^{-1}(c + A^T \lambda) - b^T \lambda$$

Properties of Quadratic Programming

1. Convexity:
   a. The problem is convex if Q is positive semidefinite. This ensures the quadratic function curves upwards and has a single minimum. If Q is not positive semidefinite, the problem may not be convex, and global optimality is not guaranteed.
2. Strong Duality:
   a. Similar to linear programming, strong duality often holds in quadratic programming when the problem is convex. This means the optimal values of the primal and dual problems are equal.
3. Applications:
   a. Quadratic programming is widely used in portfolio optimization, where the objective is to minimize risk (represented by the quadratic term) subject to achieving a certain level of return (represented by the linear term).
   b. In machine learning, Quadratic Programming is used in support vector machines (SVMs) to find the optimal separating hyperplane that maximizes the margin between two classes.

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